SZABIST
University

Shaheed Zulfikar Ali Bhutto
Institute of Science and Technology

ROBOTICS & ARTIFICIAL INTELLIGENCE DEPARTMENT

Total Marks:_____

Obtained Marks: _____

Computing Vision

Lab Task # 02

Submitted To: **Mr. Muhammad Azhar**

Student Name: **M. Mateen Bashir**

Reg. Number: **23108125**

Windows Python Environment Install and Setup

Step 1: Download and Install Python

Installing Anaconda Navigator

Create a Jupyter Notebook File (.ipynb)

Basic Operation on Jupyter Notebook

Solution:

Windows Python Environment Setup Guide

This repository provides a step-by-step guide for installing and configuring Python, Anaconda, and Jupyter Notebook on Windows.

The guide is designed for beginners and university lab environments to ensure proper setup before starting Python development and data science projects.

Step 1: Install Python

1. Download the latest version from: https://www.python.org/downloads/
2. During installation:
    - Check Install for all users
    - Check Add Python to PATH
    - Click Install Now
3. Approve installation permissions when prompted.
4. Disable the Path Length Limit to avoid directory path errors (important if path exceeds 260 characters).
5. Close the installer once completed.

Step 2: Install Anaconda Navigator

1. Download from: https://www.anaconda.com/download
2. Click Download (Skip Registration if prompted).
3. Follow installation steps:
    - Click Next
    - Accept License Agreement
    - Select Just Me installation

    - Choose installation directory (avoid spaces & Unicode characters)
    - Select Advanced Options (register as default Python if required)
    - Click Install
4. Finish installation when complete.

Step 3: Launch Jupyter Notebook

1. Open Anaconda Navigator
2. Locate Jupyter Notebook
3. Click Launch
4. Select your desired directory
5. Click New → Python 3
6. A new .ipynb notebook file will open

Tools Overview

Python

General-purpose programming language used for development and data analysis.

Anaconda

Distribution for scientific computing that includes:

- Jupyter Notebook
- Pre-installed libraries
- Package management tools

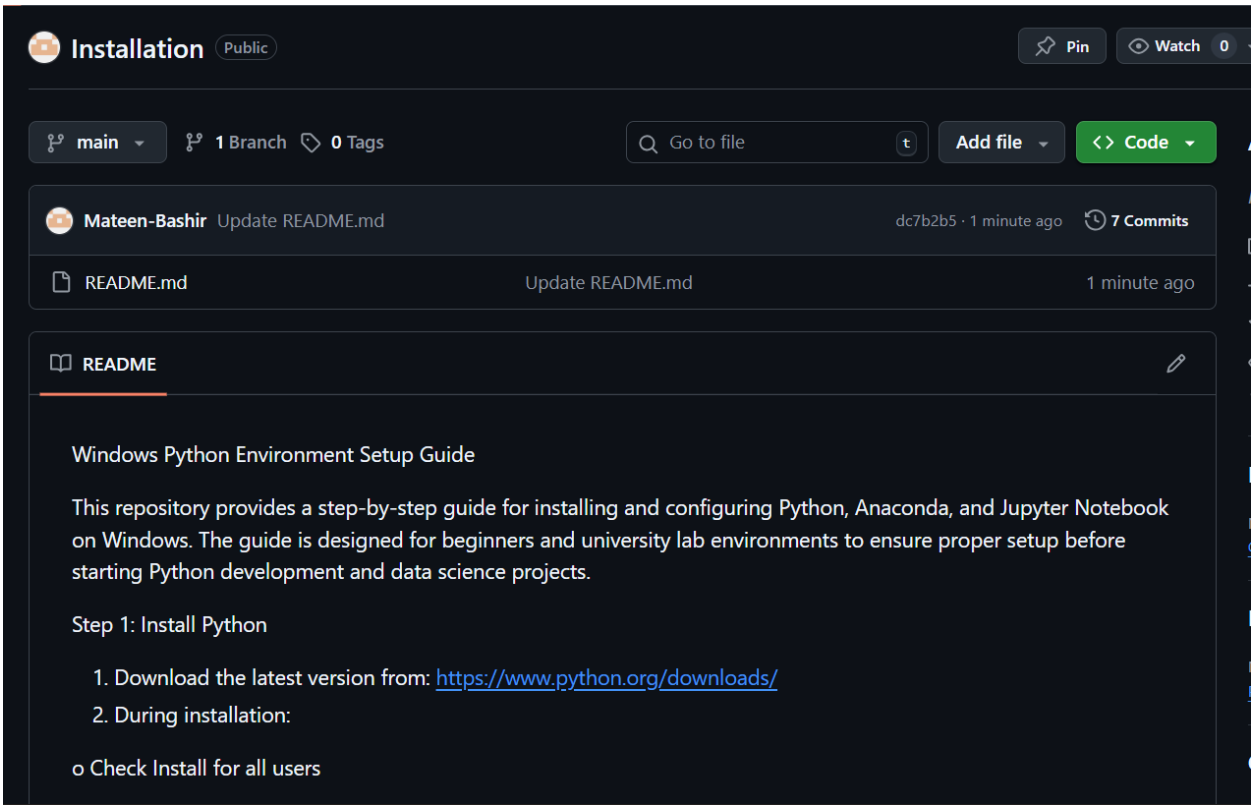
PyCharm

Professional IDE focused on software development with debugging and refactoring tools.

Purpose

This setup ensures:

- Proper Python configuration
- Correct environment variable setup
- Access to Jupyter Notebook
- Compatibility for data science and machine learning tasks



https://github.com/Mateen-Bashir